

Behind the Packets: Wireshark Analysis Report- from ICMP to TLS

Behind the Packets: A SOC Analyst's Detailed Protocol Analysis Report Using Wireshark (Days 6–10) – From ICMP Echoes to TLS Encryption Metadata.

Analyst : LAO-TSE

Tool : WIRESHARK

Focus : Network Traffic Analysis - Protocol Behaviour and Threat Indicators.

Protocols Analyzed:

Day 6 : ICMP – Echo Requests, Replies, and Destination Unreachables

Day 7 : TCP – Handshakes, Connection Teardowns, and RST Flags

Day 8 : HTTP – Unencrypted Web Requests and Session Data

Day 9 : TLS – Handshake Metadata, Certificate Details, and SNI

Day 10 : Packet Filtering Techniques – Display & Capture Filters for Targeted Traffic

Day 6: Intro to Wireshark – Packet Analysis for SOC Analysts

Objective:

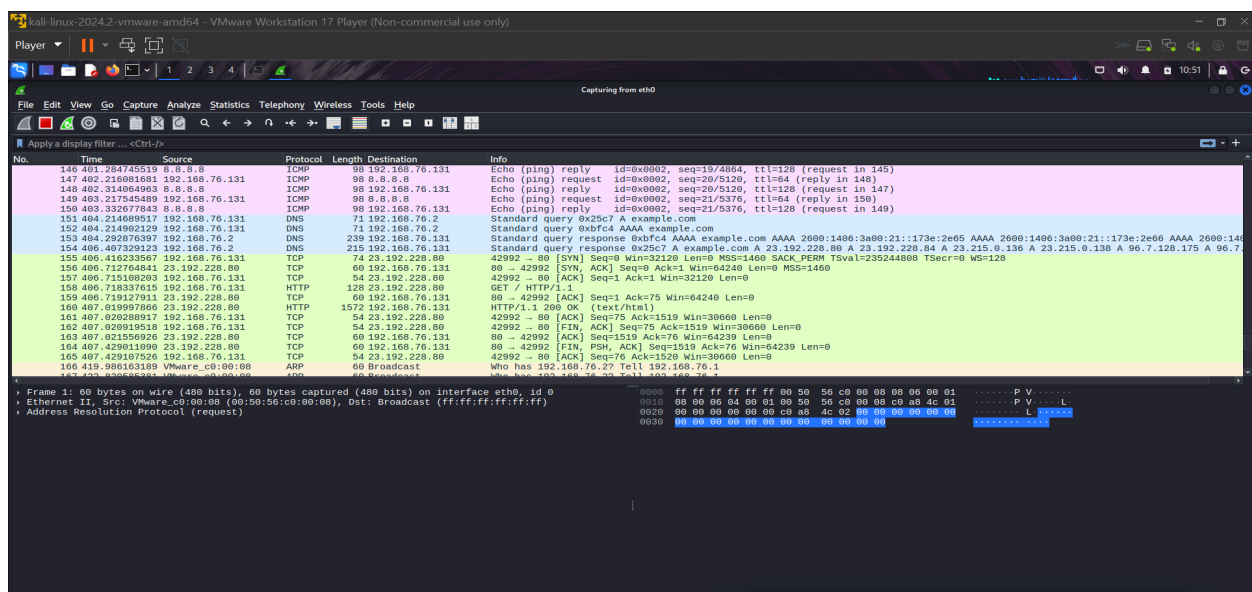
Gain foundational experience in capturing and analyzing packets using Wireshark, with a focus on identifying basic ICMP activity such as tunneling, ping sweeps, and C2 communications.

Tasks Performed:

Created a custom profile named "SOC Analyst" to streamline workflow

Applied Display Filter: icmp to isolate ICMP echo request/reply

Applied Capture Filter: icmp to collect ICMP packets during live capture



Key Takeaways:

ICMP traffic can be an indicator of both legitimate diagnostics and early-stage malicious behavior

Custom Wireshark profiles improve investigative efficiency

Capture vs Display filters serve different purposes in packet analysis

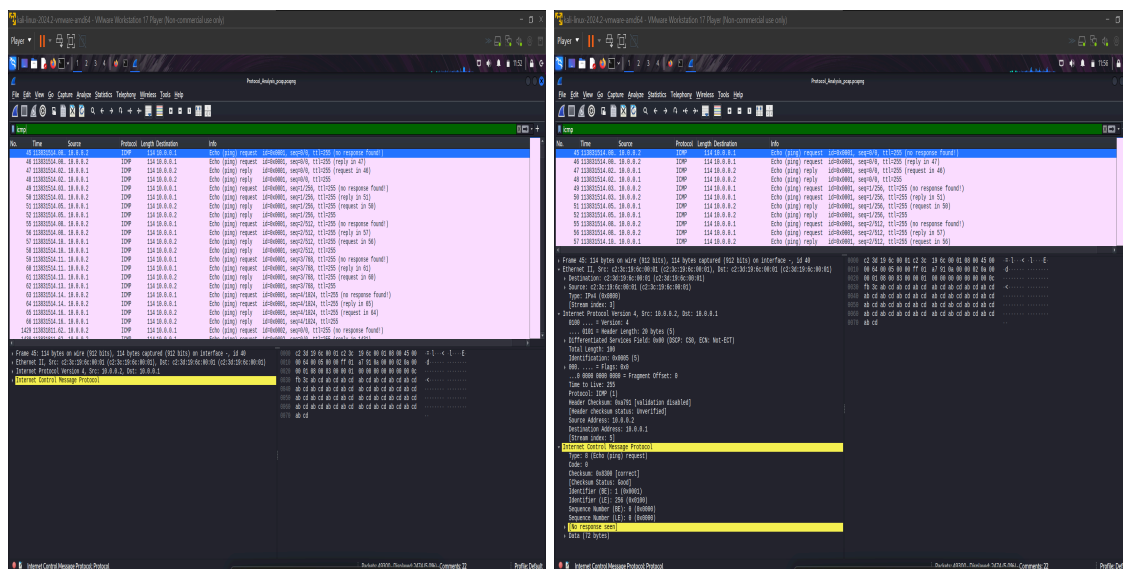
Day 7: Wireshark & the Silent Signals – ICMP Analysis

Objective:

Deepen understanding of ICMP packet types and their role in network diagnostics and reconnaissance activities.

Tasks Performed:

Analyzed icmp.type == 8 (Echo Request) and icmp.type == 0 (Echo Reply)
Detected Destination Unreachable messages for blocked routes
Matched Echo Requests and Replies using ID and sequence number



Key Takeaways:

A surge in Type 8 packets with no reply may indicate host discovery attempts.

Destination Unreachables can reveal firewall rules or misconfigured routing.

ICMP remains a valuable source of insight, especially when used quietly by threat actors.

Day 8: TCP Protocol Analysis with Wireshark

Objective:

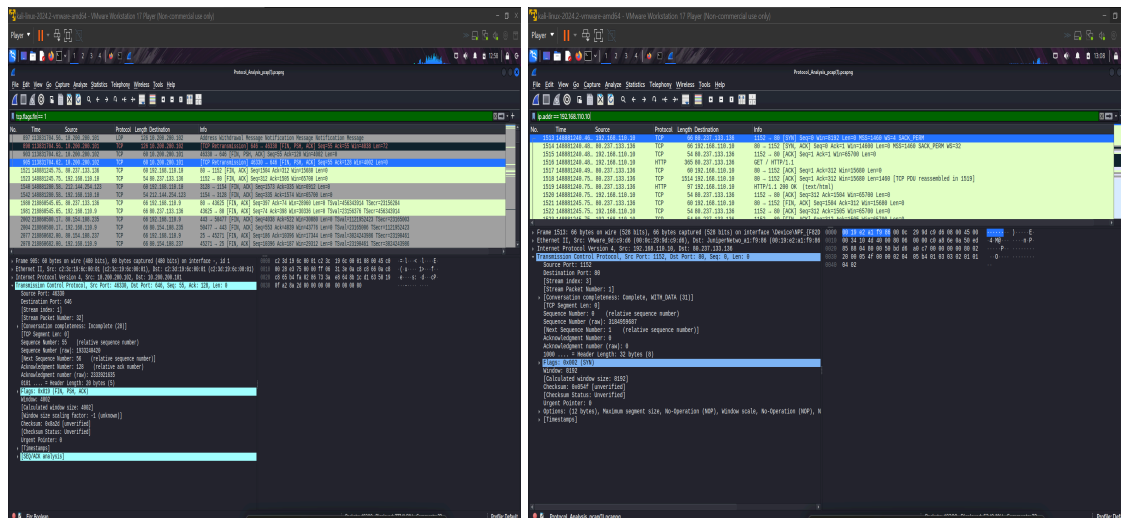
Learn to monitor and interpret TCP handshakes and connection states to detect anomalies and possible scanning behavior.

Tasks Performed:

Observed TCP 3-way handshake using `tcp.flags.syn == 1`

Identified connection teardown via `tcp.flags.fin == 1` and resets with `tcp.flags.reset == 1`

Focused on HTTP traffic using `tcp.port == 80`



Key Takeaways:

TCP handshakes indicate the start of communication; repeated SYNs may signal port scanning.

RST flags can indicate unexpected termination, which may point to scanning tools or blocked connections.

Every TCP flag combination offers context into session behavior.

Day 9: HTTP Protocol Analysis with Wireshark

Objective:

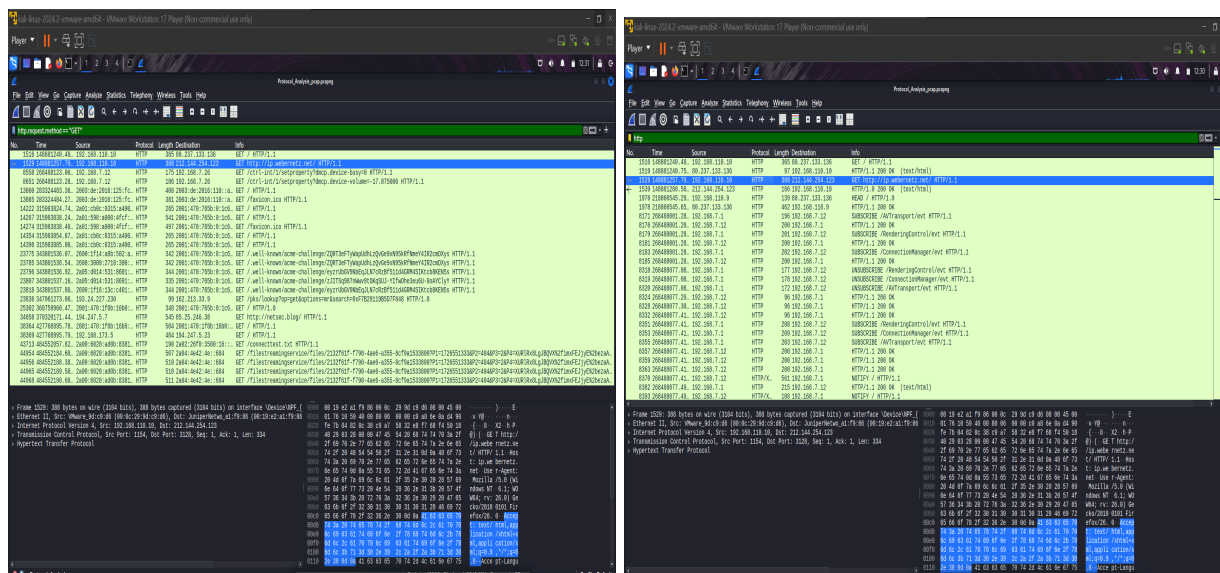
Inspect unencrypted web traffic (HTTP) to identify signs of data leakage, suspicious activity, and malware delivery.

Tasks Performed:

Reviewed packet details on HTTP port 80

Extracted headers: Host, User-Agent, URI, Status Code

Applied filters such as `http.request.method == "GET"`, `http.set_cookie`, `http.request.uri`



Key Takeaways:

HTTP is transparent and easy to analyze, making it useful for detecting unauthorized access and leaked credentials

Cookies and URIs often carry session identifiers that attackers can exploit

HTTP analysis supports early detection of C2 and malicious downloads

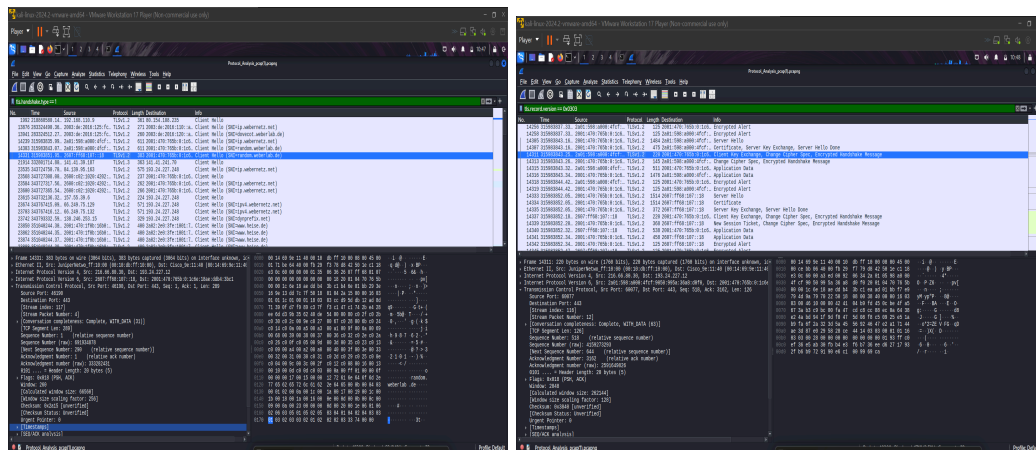
Day 10: TLS Traffic Analysis with Wireshark

Objective:

Understand how to analyze TLS traffic metadata (without decryption) to detect certificate issues and unsafe protocol versions.

Tasks Performed:

Monitored TLS handshake packets: Client Hello, Server Hello, Certificate
Used filters: `tls, tls.handshake.type == 1, tls.record.version == 0x0303`
Identified cipher suites and Server Name Indication (SNI) values



Key Takeaways:

Even without payload access, TLS metadata reveals important threat indicators (e.g., self-signed certs, outdated TLS versions).

SNI can expose the destination domain even in encrypted sessions.

TLS analysis is critical for spotting misconfigurations and malicious encryption abuse.

Summary

Across Days 6–10, I developed core Wireshark skills by focusing on specific protocols and analyzing their structure and behavioral patterns. This included:

- ICMP for diagnostic and scanning detection
- TCP for connection state monitoring
- HTTP for transparent payload analysis
- TLS for encrypted session metadata review
- Targeted filtering for focused forensic workflows
- These skills are essential for real-world SOC operations including:
- Threat detection
- Network forensics
- Incident response